



CRIMINAL FACIAL RECOGNITION USING AI ,DEEP LEARNING & MACHINE LEARNING



Team Name : Pixel Patrol

Team Members :

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FACE RECOGNITION

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Project Summary

Objective:

To develop an AI-powered **Criminal Facial Recognition System** for law enforcement.

Technology Used:

AI, Deep Learning, and Machine Learning for accurate facial identification.

Functionality:

Captures and analyzes facial features from surveillance footage.

Matches detected faces against a criminal database.

Triggers **instant SMS and sound alerts** when a match is found.

Impact:

Enhances **crime detection efficiency** and **reduces investigation time**.

Provides **real-time automated suspect identification** for law enforcement.

Improves **public safety** with quick and reliable criminal recognition.



INTRODUCTION

What is Facial Recognition?

- Facial recognition is a biometric technology that identifies or verifies individuals based on their facial features.
- It analyzes patterns based on a person's facial textures and shapes.

Aim of the System

“To develop an intelligent system capable of recognizing criminals using AI & ML algorithms.”



PROBLEM STATEMENT

- Increasing crime rates require faster and more accurate identification methods.
- Manual identification is time consuming, slow, error-prone, and inefficient in fast paced environments.
- Developing an AI – ML powered system for real time , accurate criminal identification to enhance public safety and assist law enforcement.



OBJECTIVES AND SOLUTION OVERVIEW

- To detect and recognize faces of criminals from images or videos.
- To integrate AI/ML for high accuracy recognition.
- To assist law enforcement in reducing crime
- Centralized criminal records platform
- Facial recognition integration
- Real time SMS notifications
- Faster more accurate investigations

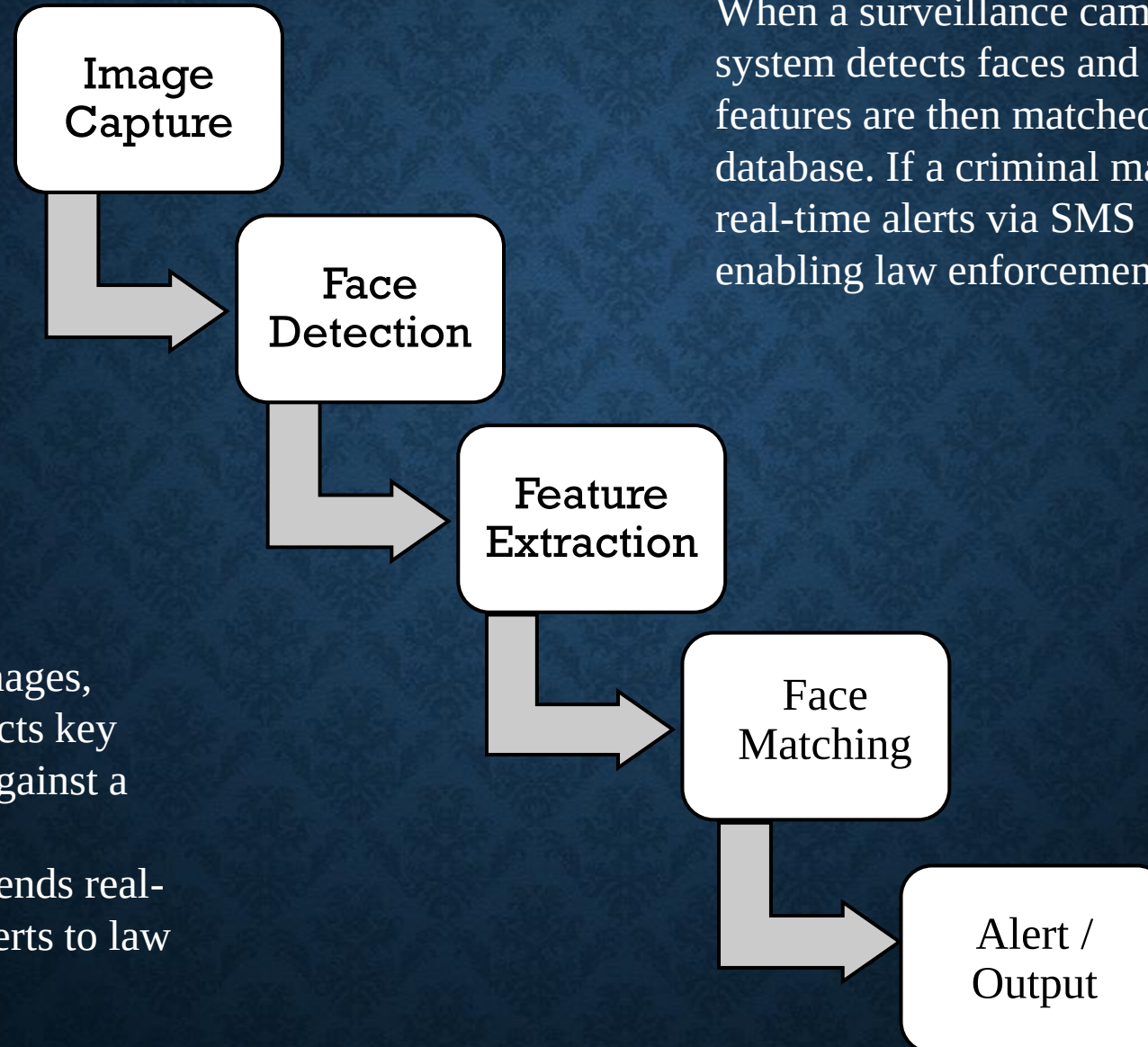


KEY FEATURES

- Real – Time Facial Detection
- High Accuracy Recognition
- Criminal Database Matching
- Automated Alerts
- Scalability
- Emotion & Behavior Analysis
- Data Logging & Reporting



When a surveillance camera captures an image, the system detects faces and extracts key features. These features are then matched against a centralized criminal database. If a criminal match is found, the system triggers real-time alerts via SMS and sound notifications, enabling law enforcement to respond immediately.



The system captures images, detects faces, and extracts key features for matching against a criminal database. If a match is found, it sends real-time SMS and audio alerts to law enforcement.



HOW IT WORKS – STEP-BY-STEP PROCESS

- Image / Video Input
- Face Detection(Haar cascade, CNN, etc.)
- Feature Extraction(Landmarks, embeddings)
- Face Matching(using ML classifiers like SVM, KNN or Deep Learning like CNNs)
- Output: Identified Person/ Criminal Database Match



TECHNOLOGY STACK / TECH STACK (TOOLS BEHIND THE SOLUTION)

- Built using Python
- Integrates key libraries & APIs
- **Frontend** : Tkinter GUI
- **Backend** : Python logic
- **Database** : SQLite
- **APIs** : SMS API
- **Other** : OpenCV and Playsound





TECHNICAL CHALLENGES

- Facial recognition accuracy
- Efficient database management
- Real – time SMS integration
- Performance optimization



INNOVATION & IMPACT

Innovation Aspects

- Real – time facial recognition & SMS integration.
- Immediate alerts for faster response
- Unique combination of technologies
- Adaptive Learning Algorithms
- Privacy & Ethical safeguards

Impact

- Enhanced Public Safety
- Reduced Workload for Officers
- Improved trust in the Justice System
- Faster Case Resolutions
- Cost – Effective Law Enforcement
- Deterrence of Criminals Activities.



TARGET USERS & BENEFITS

- Target Users

1. Law Enforcement
2. Government agencies
3. Security firms

- Potential Benefits

1. Faster suspect identification
2. Improved accuracy
3. Streamlined operations



IMPLEMENTATION PLAN

Phase	Timeframe	Deliverables	Team Members
Research, Wireframe & Design	2 Days	Wireframes, Design Documents	Deeksha Raj & Arshad Pasha
Core Development	6 Days	Functional Code, Initial Version	Arshad Pasha & Rahitya
Testing & Iteration	1 Day	Bug Reports, Improvements	Deeksha Raj
Final Polish & Demo Prep	3 Days	Refined Code, Demo Presentation	Arshad Pasha & Rahitya



FUTURE SCOPE

- Integration with smart city surveillance with Live
- AI based behaviour analysis
- Emotion detection
- Multi – modal biometric systems
- E.G:- Voice + Face + Infrared Sensors



DEMO OR PROTOTYPE

<https://youtu.be/0lMsasNGoFA?si=YgLsLk0gU8VAVbxO>

"The above link demonstrates the design and operation of our Criminal Facial Detection System."